

Introduction & Background

The goal of our project is to investigate using multi-agent robotic systems for the purpose of search and rescue operations. Search and rescue missions can incur high costs in time and resources, but automation can address these issues.

Other Solutions

While we observed that there were other possible approaches to our problem, we noted that they seemed to be less effective. We ruled out using an Advantage Actor algorithm, as we found that these algorithms had worse performance in comparison to MAPPO algorithms. We also could not use a Deep Q Network, as these algorithms only apply with discrete action spaces.

Novelty & Impact

The novelty of our project is in the combination of various aspects in the scenario of search and rescue. There is very limited research on using a Multi-Agent Reinforcement Learning (MARL) framework with Graph Neural Networks (GNN). The impact of our project is quite clear; we aim to improve search missions that could save lives.

Methods

For our MARL framework, we used Multi-Agent Proximal Policy Optimization. This involves training with global information and both shared and global

rewards. We then wrote a custom GNN that computes actions for each agent using shared information, seen in Figure 3. We implemented our environment through the Petting Zoo library and the training through Stable Baselines 3. We also trained a YOLOv11 for target detection. The software was tailored for the HiWonder JetAuto as seen in Figure 1 and Figure 2.

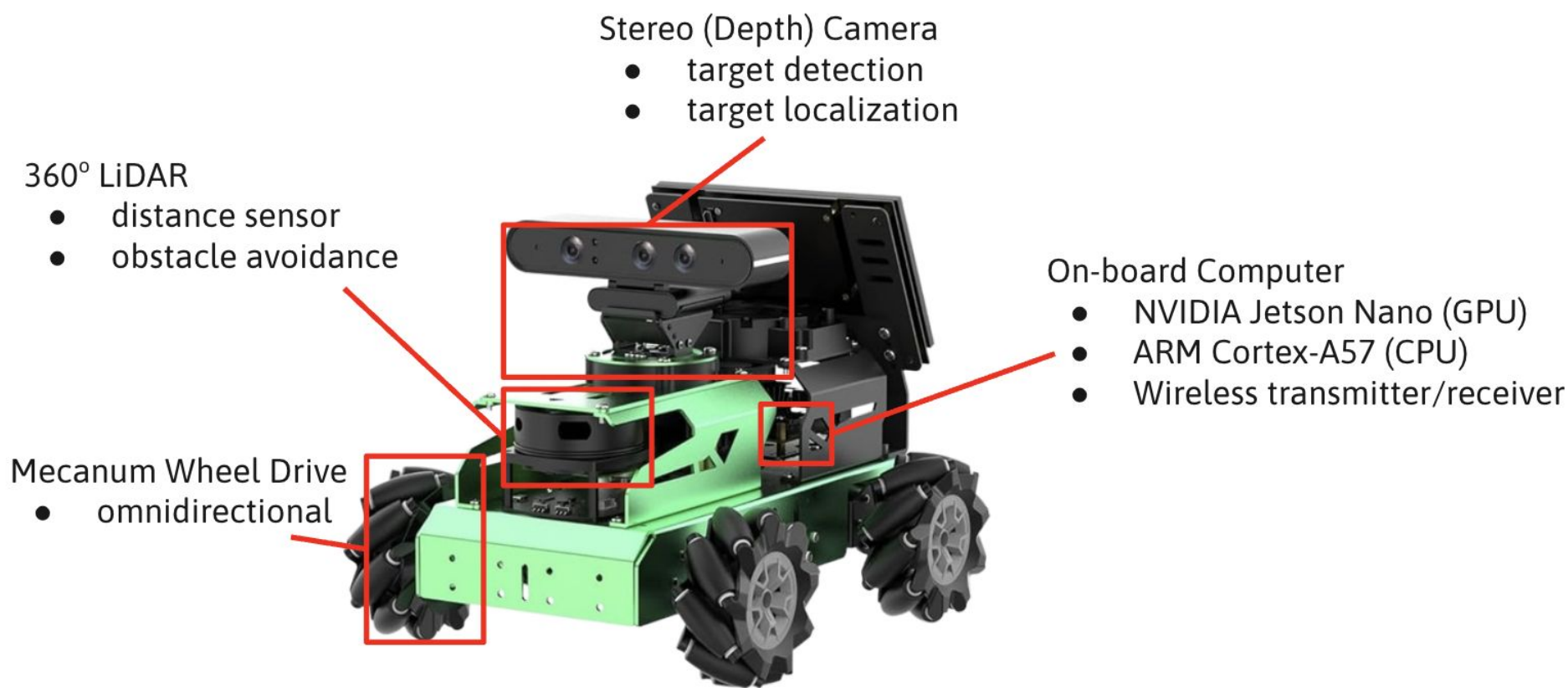


Figure 1: Capabilities of the HiWonder JetAuto.

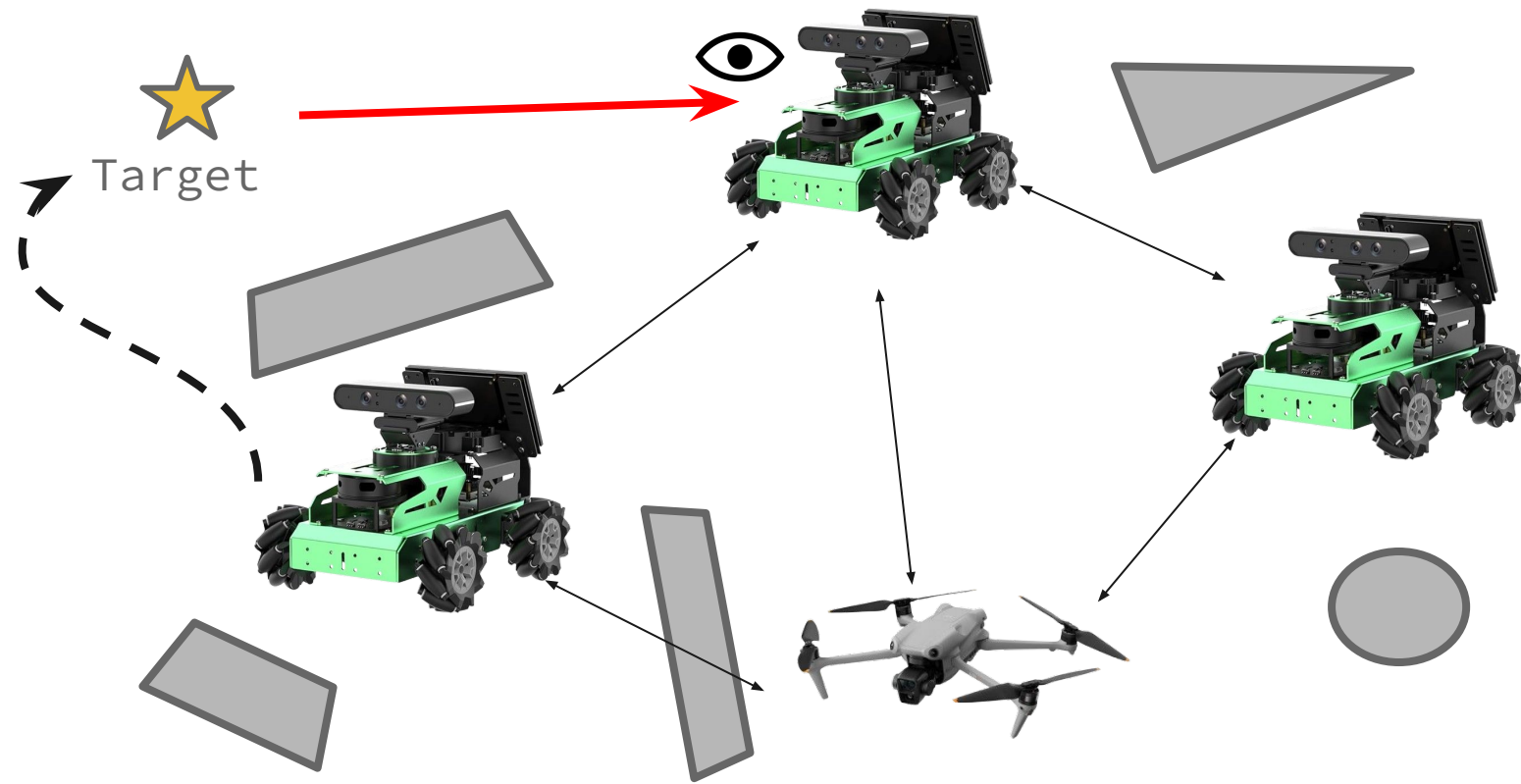


Figure 2: The application of our project as we envisioned.

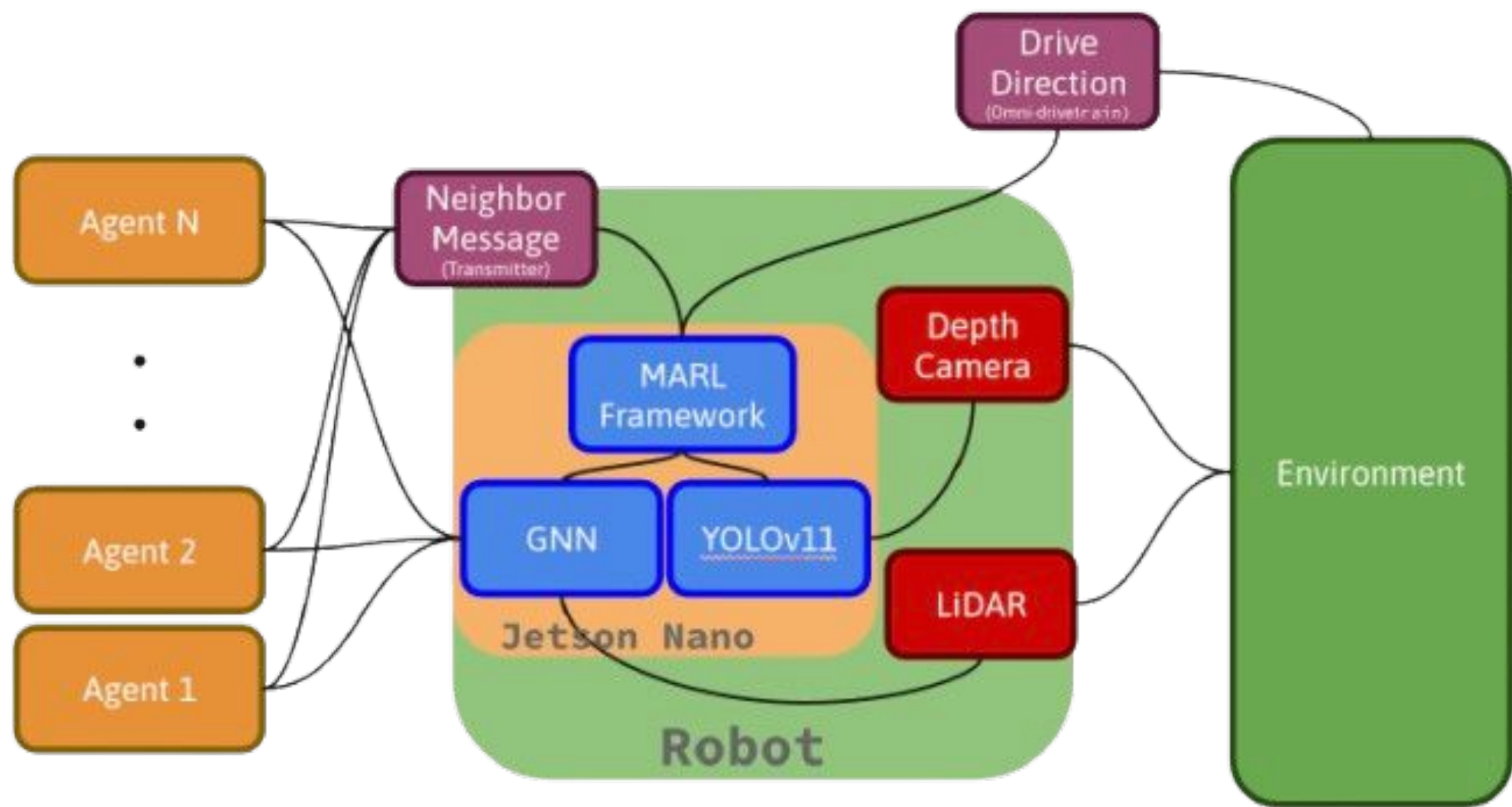


Figure 3: Diagram showing each of the components involved in the project and how they are related.

Results

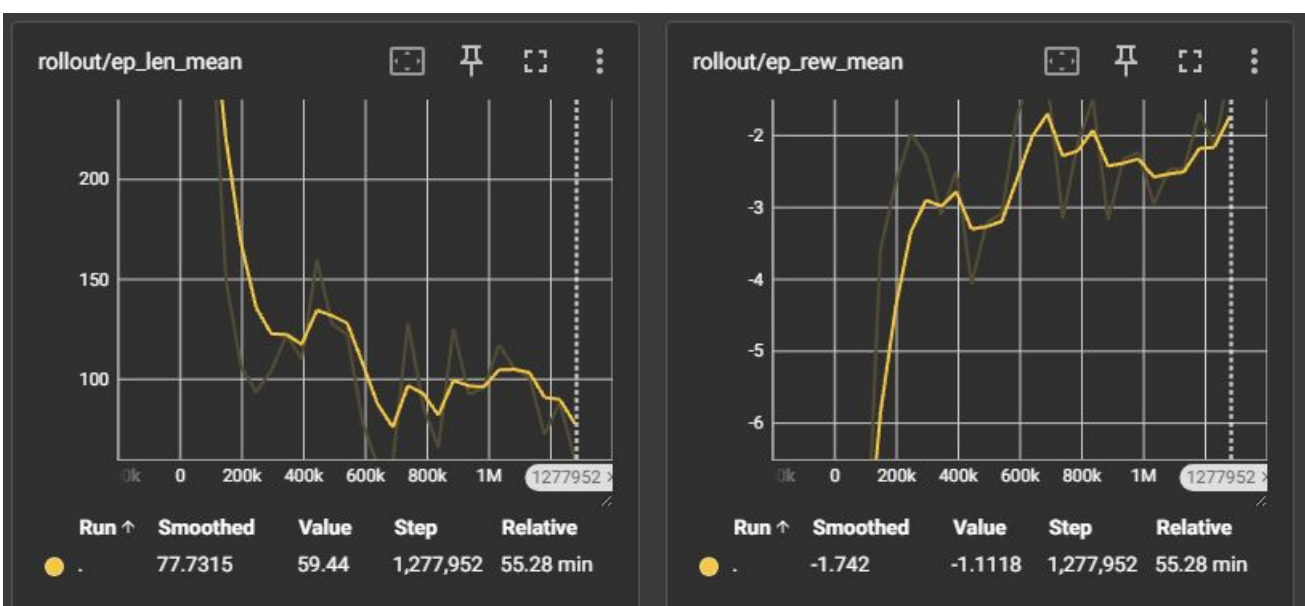


Figure 4. Graphs of the average steps and reward throughout training.

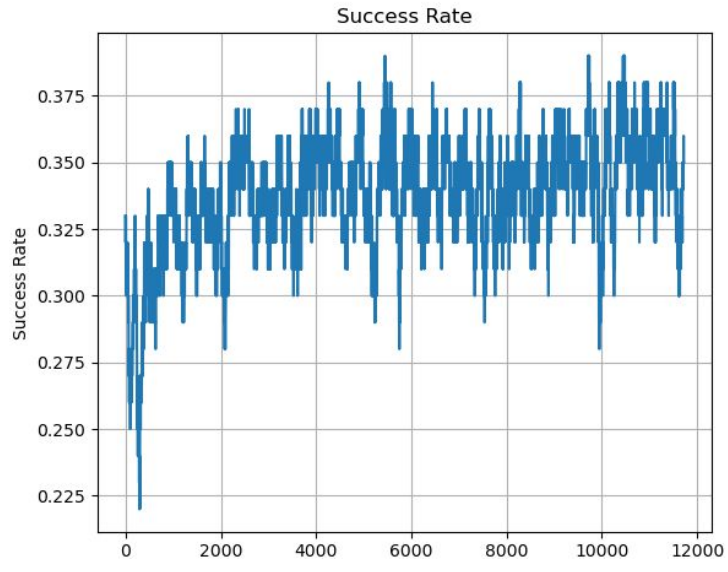


Figure 5. Graph of the success rate throughout training.

As seen in Figure 4, the average number of steps dropped by 80%. Additionally, we observed the results to be generalizable to more agents, obstacles, and a larger environment.

Conclusion

We have developed a novel MARL framework for searching, using GNNs, LiDAR, and stereoscopic object detection to maximize ground-based and aerial search efficiency. Additionally, we have demonstrated the effectiveness of our algorithm through testing our algorithm through the physical agents in a randomly constructed environment.

Further Research

In the future, we hope to incorporate the physical robots to test if our theoretical algorithm is successful. Some other ideas that we currently have include incorporating a 3rd dimension of height, using optical inputs for greater generalization, and manipulating the parameters such as the number of agents to observe their effect. Additionally, some other ideas include using a value Decomposition Network for training, or using a meta learning algorithm for optimization of hyperparameters.